

Class Activity 2 - MC

NAME:N.AKSHAYA

REG.NO:192425129

COURSE CODE:MLA0305

COURSE NAME:REINFORCEMENT LEARNING

R-CODE:

MONTE CARLO METHOD - REAL TIME EXAMPLE

Example: Daily Coffee Shop Profit Estimation

One Complete R Program

Clear Console

```
cat("\014")
```

```
set.seed(123)
```

REAL-TIME DATA

```
days <- paste("Day",1:10)
```

```
daily_profit <- sample(seq(450,900,50),10,replace=TRUE)
```

PROCESS 1 : EPISODE

```
cat("=====\n")
```

```
cat("PROCESS 1 : EPISODE\n")
```

```
cat("=====\n\n")
```

```
cat("Explanation:\n")
```

```
cat("Each episode represents one working day in a coffee shop.\n")
```

```
cat("Every day has a different profit depending on customer visits.\n\n")
```

```
episode_colors <- rainbow(length(daily_profit))
```

```
bp <- barplot(daily_profit,
```

```

        names.arg=days,
        col=episode_colors,
        border="black",
        main="Episode: Daily Coffee Shop Profit",
        xlab="Working Days",
        ylab="Profit (Rs)",
        ylim=c(0,max(daily_profit)+150))
text(bp,daily_profit,
     labels=daily_profit,
     pos=3,
     cex=0.8)
readline("Press ENTER to Continue...")
# PROCESS 2 : COLLECT REWARDS
cat("\014")
cat("=====\n")
cat("PROCESS 2 : COLLECT REWARDS\n")
cat("=====\n\n")
cat("Explanation:\n")
cat("The profit earned each day becomes the reward collected\n")
cat("by the Monte Carlo agent.\n\n")
reward <- daily_profit
reward_colors <- heat.colors(length(reward))
bp <- barplot(reward,
              names.arg=days,
              col=reward_colors,
              border="black",

```

```

        main="Collect Rewards (Daily Profit)",
        xlab="Working Days",
        ylab="Reward (Profit)",
        ylim=c(0,max(reward)+150))
text(bp,reward,
     labels=reward,
     pos=3,
     cex=0.8)
readline("Press ENTER to Continue...")
# PROCESS 3 : COMPUTE RETURN
cat("\014")
cat("=====\n")
cat("PROCESS 3 : COMPUTE RETURN\n")
cat("=====\n\n")
cat("Explanation:\n")
cat("Monte Carlo computes the return.\n")
cat("Here the return is the cumulative profit earned up to\n")
cat("each working day.\n\n")
returns <- cumsum(reward)
return_colors <- terrain.colors(length(returns))
bp <- barplot(returns,
              names.arg=days,
              col=return_colors,
              border="black",
              main="Compute Return (Cumulative Profit)",
              xlab="Working Days",

```

```

        ylab="Cumulative Profit",
        ylim=c(0,max(returns)+300))
text(bp,returns,
     labels=returns,
     pos=3,
     cex=0.8)
readline("Press ENTER to Continue...")
# PROCESS 4 : UPDATE VALUES
cat("\014")
cat("=====\n")
cat("PROCESS 4 : UPDATE VALUES\n")
cat("=====\n\n")
cat("Explanation:\n")
cat("Monte Carlo updates the estimated value.\n")
cat("Here we estimate the average daily profit after each day.\n\n")
value_estimate <- cumsum(reward)/(1:length(reward))
value_colors <- topo.colors(length(value_estimate))
bp <- barplot(value_estimate,
              names.arg=days,
              col=value_colors,
              border="black",
              main="Update Values (Average Profit Estimate)",
              xlab="Working Days",
              ylab="Estimated Average Profit",
              ylim=c(0,max(value_estimate)+150))
text(bp,value_estimate,

```

```

    labels=round(value_estimate,1),
    pos=3,
    cex=0.8)
readline("Press ENTER to Continue...")
# PROCESS 5 : REPEAT
cat("\014")
cat("=====\n")
cat("PROCESS 5 : REPEAT\n")
cat("=====\n\n")
cat("Explanation:\n")
cat("Monte Carlo repeats the same process for many episodes.\n")
cat("As more working days are observed, the estimated average\n")
cat("profit becomes more stable.\n\n")
extra_profit <- sample(seq(450,900,50),20,replace=TRUE)
repeat_estimate <- cumsum(extra_profit)/(1:length(extra_profit))
# Different color for every episode
repeat_colors <- c(
  "red","orange","gold","yellow","yellowgreen",
  "green","springgreen","turquoise","cyan","deepskyblue",
  "dodgerblue","blue","slateblue","purple","violet",
  "magenta","hotpink","brown","coral","darkred"
)
bp <- barplot(repeat_estimate,
  col=repeat_colors,
  border="black",
  main="Repeat Process (Improving Average Profit Estimate)",

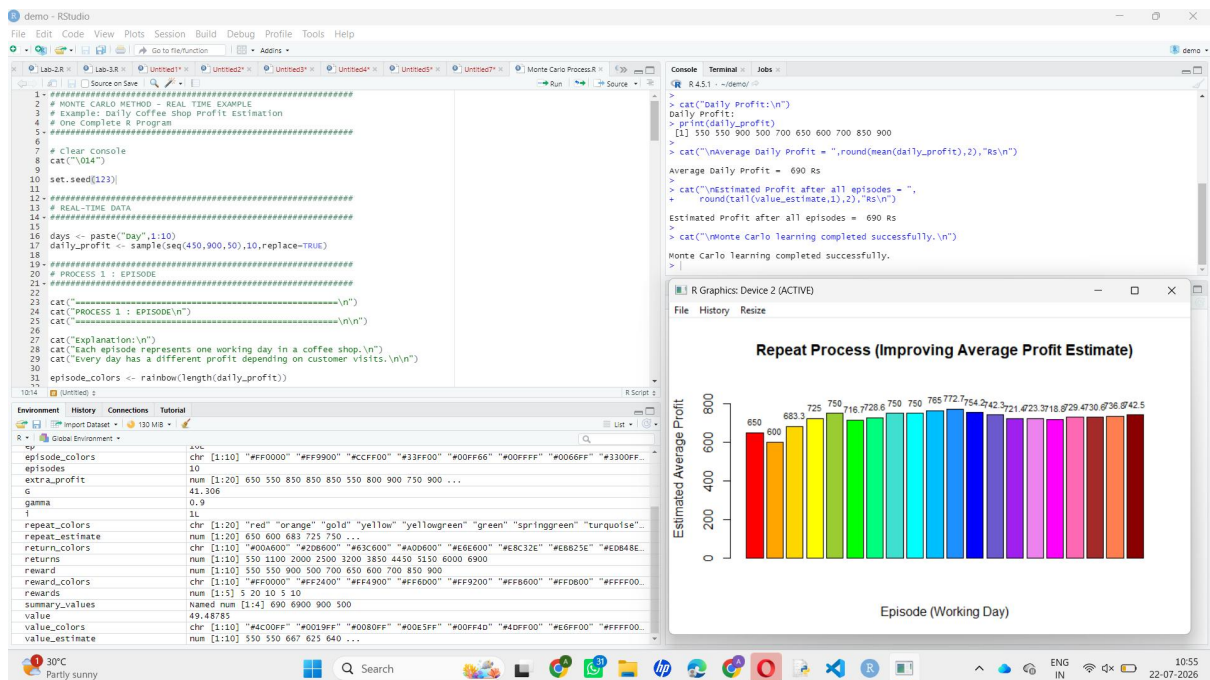
```

```

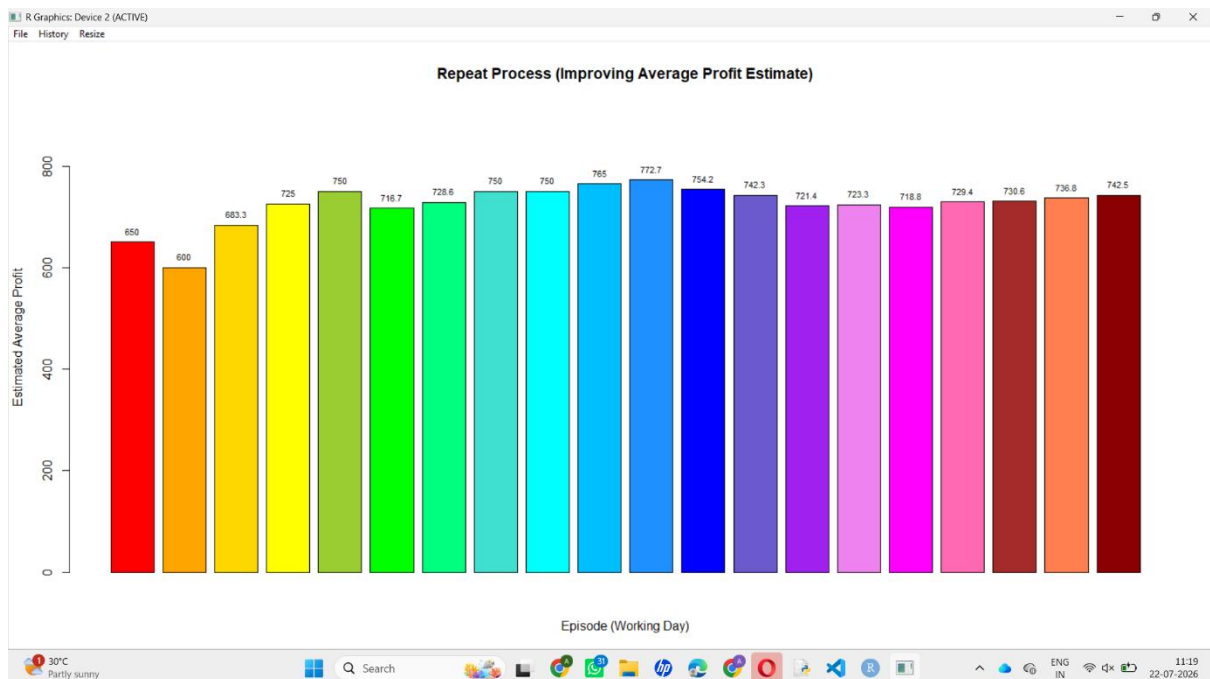
        xlab="Episode (Working Day)",
        ylab="Estimated Average Profit",
        ylim=c(0,max(repeat_estimate)+150))
text(bp,
     repeat_estimate,
     labels=round(repeat_estimate,1),
     pos=3,
     cex=0.7)
readline("Press ENTER for Final Summary...")
# FINAL SUMMARY
cat("\014")
cat("=====\n")
cat("FINAL SUMMARY OF MONTE CARLO METHOD\n")
cat("=====\n\n")
cat("Episode      : One working day in the coffee shop.\n")
cat("Collect Reward : Profit earned each day.\n")
cat("Compute Return : Cumulative profit over days.\n")
cat("Update Values  : Running average daily profit.\n")
cat("Repeat        : Observe more days to improve estimate.\n\n")
cat("Daily Profit:\n")
print(daily_profit)
cat("\nAverage Daily Profit = ",round(mean(daily_profit),2),"Rs\n")
cat("\nEstimated Profit after all episodes = ",
    round(tail(value_estimate,1),2),"Rs\n")
cat("\nMonte Carlo learning completed successfully.\n")

```

Full screenshot of Rstudio:



Visual result:



Summary result:

```
demo - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Jobs
R 4.5.1 --demo
> cat("-----\n")
> cat("FINAL SUMMARY OF MONTE CARLO METHOD\n")
FINAL SUMMARY OF MONTE CARLO METHOD
> cat("-----\n\n")

> cat("Episode      : One working day in the coffee shop.\n")
Episode      : One working day in the coffee shop.
> cat("Collect Reward : Profit earned each day.\n")
Collect Reward : Profit earned each day.
> cat("Compute Return : Cumulative profit over days.\n")
Compute Return : Cumulative profit over days.
> cat("Update Values  : Running average daily profit.\n")
Update Values  : Running average daily profit.
> cat("Repeat       : Observe more days to improve estimate.\n\n")
Repeat       : Observe more days to improve estimate.

>
> cat("Daily Profit:\n")
Daily Profit:
> print(daily_profit)
[1] 550 550 900 500 500 700 650 600 700 850 900
>
> cat("\naverage Daily Profit = ",round(mean(daily_profit),2),"Rs\n")
Average Daily Profit = 690 Rs
>
> cat("\nEstimated Profit after all episodes = ",
+   round(tail(value_estimate,1),2),"Rs\n")
Estimated Profit after all episodes = 690 Rs
>
> cat("\nMonte Carlo learning completed successfully.\n")
Monte Carlo learning completed successfully.
> |
```